

Design Ideas Notes

- ~~Requirements~~ quick to design broad idea of aircraft
- ~~Required~~ Total Kilometrage ^{be} ~~is~~ ^{max} 1130 km to reach (can't to reach Montague Airport in California)
- ~~100 HP~~ ¹⁶⁰ HP Engine under 10k
- 2 seater ^{~130}
- Top speed ~~160-200~~ knots ^{max with reserve}
- ~~55~~ gallons ^{max with reserve} to maintain around 100 knots to make 1130 km.
- Designed specifically to reduce drag to lower max ~~fuel~~ fuel capacity.
- Smaller wing area
- Low wing
- Straight wing
- ~~Longing for~~ ~~in details~~
- Wheel Pairings
- Use wing loading $Weight_{max} = \frac{1}{2} C_{Lmax} \rho S V^3$
- max weight ~~720~~ 770 kg - 860 kg
- Then to power loading $\frac{W}{HP}$ - power loading, 4.5 - 5.4 kg is good performance to 5.9 - 6.8 kg but for okay slower performance for a 160 HP
- Max weight ~~both~~ from 720 - 860 kg fit well
- 2 blade